

Clinic Appointment System

Objective

Build a clinic system where **patients book appointments**, **receptionists manage scheduling**, and **doctors manage consultations**, with correct **permissions**, **scheduling rules**, and **workflows**.

Roles

1. Patient

- Register/login
- View/update own profile
- Book available appointment slots
- View own appointments (upcoming/history)
- Cancel (with policy) + request reschedule
- View consultation summary after completion (read-only)

2. Doctor

- View own schedule + daily queue
- Confirm/decline appointment requests
- Mark checked-in/completed/no-show
- Fill consultation record (notes, diagnosis, prescription, tests)

3. Receptionist

- Manage doctor schedules (availability + exceptions)
- Confirm bookings (if clinic policy)
- Check-in patients + manage queue order

- Reschedule on behalf of patient
- Cannot access medical notes

4. Admin

- Manage users/roles
 - View analytics dashboard + export CSV
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Required Features (MUST)

1) Doctor Availability + Slot Generation

- Weekly schedule per doctor (day, start, end)
- Exceptions: vacation/day off + one-off working day
- Slot generation based on session duration (15/30 min)

2) Booking + Conflict Rules

- Patient can book **only available slots**
- Prevent:
 - Double booking for same doctor/time
 - Overlapping appointments for the same patient
- Add **buffer time** before/after (configurable, default 5 min)

3) Appointment Lifecycle

Statuses:

REQUESTED → CONFIRMED → CHECKED_IN → COMPLETED
and CANCELLED / NO_SHOW

Rules:

- Patient can cancel only if status is REQUESTED/CONFIRMED
- Doctor or receptionist can mark NO_SHOW
- COMPLETED requires consultation record completed

4) Rescheduling + Audit Trail

- Reschedule request by patient OR staff
- Must keep history:
 - old datetime, new datetime, changed by, reason, timestamp

5) Queue / Check-in

- Receptionist checks-in patients
- Doctor dashboard shows today queue ordered by check-in time
- Show waiting time (simple: now - check-in time)

6) EMR Lite (Consultation Record)

- For COMPLETED appointments:
 - diagnosis, notes
 - prescription items (drug, dose, duration)
 - requested tests
- Patient can view summary, not edit
- Receptionist cannot see medical notes

7) Search & Filters

- Appointment list filters:

- status, date range, doctor, patient (staff only)
- Search by appointment id or patient name (staff only)

General rules

- DRF API for appointments + slots
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Bonus Features

- For patients, implement login with social media FB, GOOGLE (OAuth, OAuth)
 - Waiting list + auto-fill on cancellation (hold 30 mins)
 - Payment simulation + refund policy + invoice PDF
 - Telemedicine type (online link + file sharing)
 - Analytics dashboard (no-show rate, peak hours, revenue)
 - Unit tests > 12 with coverage report
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Technical Constraints

- Use Django authentication + groups/permissions
 - Use class-based views if reasonable
 - Use database constraints to enforce unique slot booking
 - Use transactions in booking/rescheduling
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Submission

1. GitHub repo link
2. README: setup, sample users (patient/doctor/receptionist/admin), how to test main flows
3. Tests: run command + evidence (screenshots or CI)

General notes

- Group members up-to 5 per team
- Beautiful design is a must (You can use AI to do this task)
- You must complete the follow.
- Admin dashboard should be implemented (Don't use the built-in one),

Best of luck ^^

Noha Shehab



Deadline: 14 April 2026 (Send me the repo for your works)

All the members must commit their work to the repo. ()

